

LOOB HOLDING SDN BHD

Technical Assessment

Lead Full Stack Developer

Beverage App Redesign Challenge · Flutter · MySQL · AWS

DETAILS	
Role	Lead Full Stack Developer
Duration	7 calendar days from receipt of this document
Submission	GitHub repository
Prepared by	Loob Holding Sdn Bhd April 2026

1. Overview

This assessment evaluates your ability to architect, design, and build a production-grade mobile application. We are looking for evidence of strategic thinking, sound engineering decisions, and the ability to justify trade-offs. Code completeness matters, but so does how you explain your choices.

This document is confidential. Please do not share the contents publicly.

2. The Challenge

You have been given the opportunity to **redesign and rebuild the mobile app** for either **Tealive** or **Baskbear Coffee**. Pick whichever brand inspires you more. These are real Malaysian beverage brands serving millions of customers across Asia.

Reimagine the experience from the ground up. There is no correct answer. We would like to see how you think, what you prioritise, and how clearly you can justify it.

2.1 What You Are Building

A cross-platform mobile app built with **Flutter**, targeting consumers across at least two countries (e.g. Malaysia, Thailand). You are free to support more; we recommend keeping it to a manageable scope given the time constraint. The app must include at minimum these four core modules:

Module	Minimum Scope
1. Menu	Item browsing by category; multi-language names and descriptions; country-specific pricing and currency; item customisation (size, add-ons, sugar/ice level); dietary tags; availability toggles per country or outlet.
2. Ordering	Cart management; order placement (dine-in, takeaway, or delivery where applicable); order status tracking; order history; handling of peak-load scenarios.
3. Vouchers	Voucher listing and redemption; percentage and fixed-amount discounts; minimum spend and per-user usage limits; expiry logic; country-specific availability. Define and justify your voucher stacking decision.
4. Multi-Country	Country selection at onboarding; localised content (language, currency, promotions); region-specific feature flags; country-specific tax rules; timezone handling.

You are free to go beyond these four modules. Any additional features should include a brief note on why you added them. As a bonus, design your solution in a way that can be deployed to new countries quickly.

2.2 Technology Requirements

Area	Requirement
Frontend	Flutter (latest stable). Target Android + iOS. Web is optional.
State Management	Your choice (BLoC, Riverpod, GetX, etc.). Justify your pick in the README.
Backend	Your choice of language and framework (e.g. Laravel, Node.js/Express, NestJS, Go). REST or GraphQL are both acceptable. Justify your choice.
Database	MySQL (preferred). You may propose an additional databases for specific needs (e.g. Redis for caching) but MySQL must be the primary data store.
Cloud	AWS. See Section 4 for architecture requirements.
Auth	Your choice (Firebase Auth, AWS Cognito, JWT, etc.). Justify your approach.
Version Control	GitHub. Your repository must be public or shared with the hiring team.

3. Design & Architecture Justification

Your README.md must address all questions in this section.

3.1 App Design & UX

Q1. Why did you design the app this way? Walk us through your design philosophy.

Q2. How did you approach multi-country UX? What decisions did you make for localisation and regionalisation?

Q3. [Optional] What additional features did you build beyond the four required modules, and why?

3.2 Flutter Architecture

Q4. Which state management solution did you choose and why? What trade-offs did you accept?

Q5. How did you structure your Flutter project? Describe your folder structure and layer separation.

Q6. How does your app handle offline scenarios or slow connectivity, which is common in Southeast Asia?

4. Database Design

Document your MySQL schema in the README with an entity list, key relationships, and justification for your design decisions. Include migration files and seeder files in the repository.

Q7. Walk us through your core schema. How do you model a menu item with country-specific pricing, an order with its line items, and a voucher with redemption rules?

Q8. How does your schema handle multi-country data?

Q9. What indexing strategy did you apply?

Q10. How would you handle MySQL schema migrations safely in a live system with no downtime?

5. AWS Architecture & Scalability

You do not need to provision live AWS infrastructure. Provide a architecture design in your README answering the questions below.

5.1 Core Infrastructure

Q11. Describe your overall AWS architecture. Which services would you use, and how do they connect?

Q12. How would you design for high availability across multiple regions?

Q13. How would you scale the ordering service to handle a flash sale (10x normal load in under 5 minutes)?

5.2 Data, Caching & Storage

Q14. Where and how would you implement caching?

Q15. How would you manage image and media assets (menu photos, banners) at scale across multiple countries?

5.3 Observability

Q16. How would you monitor this application in production?

Q17. Describe your CI/CD pipeline and the stages it goes through.

6. Deliverables

All of the following must be present in your GitHub repository at the time of submission.

#	Deliverable	Details
1	Flutter Source Code	Complete, buildable Flutter project. Must run on Android or iOS simulator. Include an .env.example so evaluators can set up the environment.
2	README.md	App overview, tech stack with justifications, setup instructions, answers to all Q&A in Sections 3 to 5.
3	MySQL Schema	SQL migration and seeder files. Include an Entity Relationship description in the README.

8. Submission Instructions

1. Create a public GitHub repository (or invite the hiring team as collaborators to a private one).
2. Name the repository: tealive-assessment or baskbear-assessment.
3. Ensure all code, documentation, schema files, and the architecture diagram are committed.
4. Share the repo with the hiring team.

The 7-day timer begins from the date and time this document is sent to you.

If you have any questions about the scope or requirements, feel free to email us and we'll get back to you within a few hours on business days.

We won't be able to advise on whether a specific technical approach is correct, as that's part of what we're looking to evaluate.

9. README.md Preferred Structure

The following is a sample structure for your README.md.

```
# [Brand Name] App Redesign - Loob Holding Sdn Bhd Assessment

> Candidate: [Your Name] | Submission Date: [Date]

## App Overview
Brief description of the brand chosen and what you built.

## Tech Stack


| Layer    | Technology | Justification |
|----------|------------|---------------|
| Frontend |            |               |
| Backend  |            |               |
| Database | MySQL      |               |
| Auth     |            |               |
| Cloud    | AWS        |               |



## Project Structure
Explain your folder layout and architecture pattern.

## Local Setup
Step-by-step instructions to run the app locally.

## Design & Architecture Decisions (Q1 to Q6)
Answer each question from Section 3 of the assessment.

## Database Design (Q7 to Q10)
Answer each question from Section 4.
Include schema description and entity relationships.

## AWS Architecture (Q11 to Q19)
Answer each question from Section 5.
Embed or link your architecture diagram here.

## CI/CD Pipeline
Describe your pipeline stages (GitHub Actions or equivalent).

## Known Limitations & What I Would Improve
Be honest about scope cut and what you would do next.
```

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